NMEA Sentences

Understanding GNSS Data — Tutorial Notes

Preparation guide for the BIP26 NMEA Quiz | TU Dublin Tallaght Campus | June 2026

How to use these notes

These notes cover the key topics tested in the NMEA quiz — work through each section before attempting the quiz.
Actual NMEA sentence examples are shown in code boxes throughout — these match the kind of data the Waveshare LC29H produces.
Questions marked 🔍 suggest a web search for the most current or detailed information.

1. What is NMEA?

NMEA stands for National Marine Electronics Association. Originally a US marine electronics standards body, NMEA created the NMEA 0183 standard which defines a simple text-based protocol for communication between GNSS receivers and other devices.

NMEA 0183 is the universal language that virtually all GPS and GNSS receivers use to output their data. When the Waveshare LC29H sends data to the Tenstar ESP32-S3, it uses NMEA 0183 sentences.

Key characteristics of NMEA 0183:

- Plain text format — every sentence is human-readable ASCII characters
- One sentence per line — each ends with a carriage return and line feed (CR+LF)
- Standard baud rate — 9600 baud by default, though many modern receivers (including the LC29H) use 115200
- Checksum — every sentence ends with a two-character hexadecimal checksum for error detection

2. NMEA Sentence Structure

Every NMEA sentence follows the same basic structure:

```
$TTSSS,field1,field2,field3,...,fieldN*CC

Where:
  $        = Start character — every sentence begins with this
  TT       = Talker ID — identifies which system sent the data (GP, GA, GL, GB,
GN...)
  SSS      = Sentence type — identifies what data is in this sentence (GGA, RMC,
GSV...)
  fields   = Comma-separated data fields
  *        = Separator before the checksum
  CC       = Checksum — two hexadecimal digits
```

A real example from the LC29H:

```
$GNGGA,123519,5320.1234,N,00615.5678,W,1,08,0.9,85.4,M,48.6,M,,*47
```

3. Talker IDs — Which Constellation?

The two-letter talker ID tells you which satellite constellation the data comes from:

Talker ID	Constellation	Example sentence
GP	GPS (USA)	$GPGSV — GPS satellites in view
GA	Galileo (EU)	$GAGSV — Galileo satellites in view
GL	GLONASS (Russia)	$GLGSV — GLONASS satellites in view
GB	BeiDou (China)	$GBGSV — BeiDou satellites in view
GQ	QZSS (Japan)	$GQGSV — QZSS satellites in view

Talker ID	Constellation	Example sentence
GN	Multiple / combined	$GNGGA — fix using multiple constellations

Key point for the quiz
GN means combined multi-constellation data — this is the most common talker ID seen from modern receivers like the LC29H.
GA specifically means Galileo — used in $GAGSV sentences which the BIP26 Galileo Tracker parses.

4. The Most Important NMEA Sentences

4.1 GGA — GPS Fix Data

GGA is the most commonly used sentence. It contains the core position fix information.

```
$GNGGA,123519,5320.1234,N,00615.5678,W,1,08,0.9,85.4,M,48.6,M,,*47
       ^^^^^^  ^^^^^^^^^^^  ^  ^^^^^^^^^^^  ^  ^^ ^^^  ^^^^  ^ ^^^^  ^
       Time    Latitude     N  Longitude    W  Q  Sat  HDOP Alt M Geo M
```

Field	Example	Meaning
Time	123519	12:35:19 UTC
Latitude	5320.1234,N	53 degrees 20.1234 minutes North
Longitude	00615.5678,W	006 degrees 15.5678 minutes West
Fix Quality	1	1=GPS fix, 2=DGPS, 4=RTK fixed, 0=no fix
Satellites Used	08	8 satellites used in this fix
HDOP	0.9	Horizontal accuracy (lower = better)
Altitude	85.4,M	85.4 metres above mean sea level
Geoid separation	48.6,M	Difference between WGS84 ellipsoid and sea level

⚠ Coordinate format — common confusion
NMEA coordinates use DDMM.MMMM format — NOT decimal degrees.
5320.1234 N means 53 degrees and 20.1234 minutes North.
To convert to decimal degrees: 53 + (20.1234/60) = 53.335390 degrees North.
TinyGPSPlus converts this automatically — gps.location.lat() returns decimal degrees.

4.2 RMC — Recommended Minimum Navigation Information

RMC is a compact sentence combining position, speed, course, and date in one line. It is the minimum information needed for basic navigation.

```
$GNRMC,123519,A,5320.1234,N,00615.5678,W,005.5,054.7,040726,,,A*XX
              ^  ^                                 ^^^^^  ^^^^^  ^^^^^^
              Status  (A=Active/valid,V=Void)  Speed  Course  Date
```

- Status A = Active (valid fix), V = Void (no fix)
- Speed is in knots (1 knot = 1.852 km/h)
- Course is in degrees clockwise from True North
- Date format is DDMMYY

4.3 GSV — Satellites in View

GSV sentences describe all the satellites that the receiver can see, even if they are not being used in the fix. Because there can be many satellites, the data is spread across multiple GSV sentences.

```
$GAGSV,2,1,07,02,45,123,38,05,67,089,42,09,23,234,35,11,54,312,41*XX
```

```
        ^ ^ ^^   ^^ ^^ ^^^   ^^ -- next sat -- -- next sat -- -- next sat --
        | | |    PRN El Az SNR
        | | Total sats in view (7)
        | This is message 1 of 2
        Total messages needed (2)
```

Field	Meaning
Total messages	How many GSV sentences needed for this constellation (1-3 typically)
Message number	Which sentence this is (1, 2, 3...)
Total sats in view	Total satellites visible for this constellation
PRN	Satellite identifier number
Elevation	Degrees above horizon (0=horizon, 90=directly overhead)
Azimuth	Compass direction in degrees (0-360)
SNR	Signal to Noise Ratio in dBHz (0=not tracked, ~50=excellent)

Each GSV sentence contains up to 4 satellites. If there are 7 Galileo satellites in view, you will see two $GAGSV sentences — the first with 4 satellites, the second with 3.

4.4 GSA — GNSS DOP and Active Satellites

GSA reports which satellites are being used in the current fix and the dilution of precision values.

```
$GNGSA,A,3,02,05,09,11,14,16,20,,,,,,1.2,0.9,0.8*XX
       ^  ^  ^^  ^^  ^^  ^^  ^^  ^^  ^^ (up to 12 PRNs)  ^^^  ^^^  ^^^
       Mode Fix  -- Satellite PRNs used in fix --          PDOP HDOP VDOP
```

- Mode 1 = Manual (forced to 2D or 3D), Mode 2 = Automatic
- Fix: 1=no fix, 2=2D fix (no altitude), 3=3D fix (full position)
- PDOP = Position DOP (overall 3D accuracy)
- HDOP = Horizontal DOP (horizontal accuracy)
- VDOP = Vertical DOP (altitude accuracy)

4.5 GLL — Geographic Position

GLL provides just latitude and longitude with UTC time — a simplified position sentence.

```
$GPGLL,5320.1234,N,00615.5678,W,123519,A*XX
```

4.6 VTG — Course and Speed

VTG provides the direction of travel and speed — velocity information.

```
$GPVTG,054.7,T,034.4,M,005.5,N,010.2,K*XX
        ^^^^^  ^  ^^^^^  ^  ^^^^^  ^  ^^^^^  ^
        True   T  Mag    M  Knots  N  km/h   K
```

4.7 ZDA — Date and Time

ZDA provides full UTC date and time including local time zone offset.

```
$GPZDA,201530.00,04,07,2026,+01,00*XX
        ^^^^^^^^  ^^  ^^  ^^^^  ^^^  ^^
        UTC time  DD  MM  YYYY  +hr  min offset
```

5. The NMEA Checksum

Every NMEA sentence ends with an asterisk (*) followed by two hexadecimal digits — the checksum. The checksum is used to detect transmission errors. If the checksum does not match, the sentence should be discarded.

The checksum is calculated by XOR-ing (exclusive OR) all the bytes between the $ and the * characters (not including either). The result is expressed as two uppercase hexadecimal digits.

```
Example: $GPGGA,...,data...*4F

The checksum 4F is the XOR of all characters between $ and *.

XOR means: if the bits are different the result is 1, if the same the result is
0.
This detects any single-bit transmission error.
```

Key point for the quiz

Checksum = XOR of all characters between $ and * (exclusive of both).
Result is two hexadecimal digits (00-FF).
If checksum fails, the sentence is corrupt — discard it.
TinyGPSPlus validates checksums automatically.

6. Proprietary NMEA Sentences — Quectel PQTM

Standard NMEA sentences start with $GP, $GA etc. Manufacturers can also define their own proprietary sentences, prefixed with $P followed by the manufacturer code.
Quectel (who make the LC29H module) uses the prefix $PQTM for their proprietary sentences:

Sentence	Purpose
$PQTMVERNO	`Query firmware version and build date of the LC29H`
$PQTMCFGosnma,R	`Query whether OSNMA authentication is enabled`
$PQTMCFGosnma,W,1	`Enable OSNMA authentication`
$PQTMOSNMA	`OSNMA authentication status output`

```
Example $PQTMVERNO response:
$PQTMVERNO,LC29HAANR11A02S,2025/10/15,09:30:00*1F
           ^^^^^^^^^^^^^^^ ^^^^^^^^^^  ^^^^^^^^
           Firmware version  Build date  Build time
```

Web search suggestion

Search for 'Quectel LC29H PQTM protocol specification' for the full list of proprietary commands supported by the LC29H module.

7. DOP — Dilution of Precision

DOP is a number that describes how well the visible satellites are geometrically positioned to give an accurate fix. A low DOP means the satellites are spread out across the sky — ideal. A high DOP means they are clustered together — less accurate.

DOP Value	Rating	Meaning
< 1	`Ideal`	Highest possible accuracy
1 - 2	`Excellent`	Position suitable for almost all applications
2 - 5	`Good`	Suitable for most applications
5 - 10	`Moderate`	Accuracy is reduced — use with caution
> 10	`Poor`	Very low accuracy — receiver may not give a fix

8. NMEA in Arduino — TinyGPSPlus

In the BIP26 sketches we use the TinyGPSPlus library to parse NMEA sentences automatically. You feed it one character at a time using gps.encode(c) and it handles all the parsing internally.

```
// Feed GPS characters into TinyGPSPlus
while (GPSSerial.available()) {
    gps.encode(GPSSerial.read());
}

// Then access the parsed data:
gps.location.lat()       // Latitude in decimal degrees
gps.location.lng()       // Longitude in decimal degrees
gps.altitude.meters()    // Altitude in metres
gps.speed.kmph()         // Speed in km/h
gps.course.deg()         // Course in degrees
gps.time.hour()          // UTC hour
gps.satellites.value()   // Number of satellites used
gps.hdop.hdop()          // HDOP value
gps.charsProcessed()     // Total characters received (useful for debugging)
```

If gps.charsProcessed() stays at zero, the GPS is not sending data — check wiring. If it counts up but gps.location.isValid() stays false, the receiver is working but has not got a fix yet — give it more time and check the antenna.

9. Questions Requiring Web Searches

- Current NMEA 0183 standard version — search 'NMEA 0183 standard version 2026'
- NMEA 2000 (the newer CAN-bus based standard) — search 'NMEA 2000 vs NMEA 0183'
- Quectel LC29H full PQTM command list — search 'Quectel LC29H protocol specification'
- TinyGPSPlus library documentation — search 'TinyGPSPlus Arduino library GitHub'

10. Glossary

Term	Definition
NMEA	National Marine Electronics Association – created the NMEA 0183 serial data standard
NMEA 0183	The serial communication standard used by virtually all GNSS receivers
Talker ID	Two-letter prefix identifying the source constellation (GP, GA, GL, GB, GN)
GGA	Global positioning system Fix dAta – main position sentence
RMC	Recommended Minimum navigation information – compact position+velocity sentence
GSV	Satellites in View – lists all visible satellites with elevation, azimuth, and SNR
GSA	GNSS DOP and Active Satellites – lists satellites used in fix and DOP values
GLL	Geographic Latitude and Longitude – simple position sentence
VTG	Velocity made good and course over ground
ZDA	Time and Date – UTC time with time zone information
HDOP	Horizontal Dilution of Precision – horizontal accuracy quality indicator
VDOP	Vertical Dilution of Precision – altitude accuracy quality indicator
PDOP	Position Dilution of Precision – overall 3D accuracy quality indicator
SNR	Signal to Noise Ratio – signal quality in dBHz (higher = stronger signal)
PRN	Pseudo-Random Noise number – unique identifier for each GNSS satellite

Term	Definition
Checksum	XOR of all sentence bytes between $ and * — used for error detection
XOR	Exclusive OR — bitwise operation used in NMEA checksum calculation
Baud rate	Serial communication speed — LC29H uses 115200 baud with Arduino
Fix	A valid calculated position from sufficient satellite signals
TinyGPSPlus	Arduino library that parses NMEA sentences automatically
PQTM	Quectel proprietary NMEA sentence prefix for the LC29H module
$PQTMVERNO	Quectel command to query LC29H firmware version

11. Reference Web Links

- NMEA 0183 Wikipedia overview: https://en.wikipedia.org/wiki/NMEA_0183
- TinyGPSPlus library: https://github.com/mikalhart/TinyGPSPlus
- NMEA sentence reference (gpsinformation.org): https://www.gpsinformation.org/dale/nmea.htm
- Quectel LC29H product page: https://www.quectel.com/product/gnss-lc29h
- Waveshare LC29H wiki: https://www.waveshare.com/wiki/LC29H(XX)_GPS/RTK_HAT
- NMEA checksum calculator (online): https://nmeachecksum.eqth.net